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Studierichting: Informatica
Oefeningenreeks 2E nr. 18.10

$$4 + \log_x \sqrt{2} - \log_x 4\sqrt{32} = 0$$

$$\Leftrightarrow 4 + \log_x \frac{\sqrt{2}}{4\sqrt{32}} = 0$$

$$\Leftrightarrow 4 + \log_x \frac{2^{\frac{1}{2}}}{2^2\sqrt{2^5}} = 0$$

$$\Leftrightarrow 4 + \log_x \left(\frac{2^{\frac{1}{2}}}{2^{\frac{9}{2}}} \right) = 0$$

$$\Leftrightarrow 4 + \log_x (2^{-\frac{8}{2}}) = 0$$

$$\Leftrightarrow 4 + \log_x 2^{-4} = 0$$

$$\Leftrightarrow \log_x 2^{-4} = -4$$

$$\Leftrightarrow x^{-4} = 2^{-4}$$

$$\Leftrightarrow x = 2$$

References